RESEARCH IMPLEMENTATION RECORD

Vehicle Tracking + Occlusions Research Project

Phase 1F

Online loss detection baselines: L1 simple rule and L2 Sentinel-inspired reliability states

Purpose of this document

A continuation record of the completed Phase 1D/1E baseline-measurement work. It documents why online loss detection was needed next, how L1 and L2 were designed, the exact runtime signals and thresholds used, the diagnostic failures that caused redesigns, the final 24-run comparisons, and the scientific conclusions that should be carried into Phase 1G candidate generation and the final research paper.

RECORD	STATE AS OF 05 OCTOBER 2026
Previous record	Phase 1D/1E completed: event-aware reference loss/recovery metrics and annotation-backed candidate-retention analysis
Phase 1F scope	Two online loss-detection baselines: L1 simple rule and L2 Sentinel-inspired reliability-state rule
Evaluation set	24 tracker-event runs = 12 frozen UA-DETRAC pilot events x SAM2.1 Tiny / DAM4SAM Tiny
Main finding	L1 is conservative; L2 catches smooth ambiguity at the risk-trigger level but introduces two false Critical triggers
Handoff	CRITICAL can trigger candidate/recovery processing; LOST remains the stronger explicit tracking-failure state

Source basis. Previous Phase 1D/1E record; generated Phase 1F CSV/terminal outputs; tracker adapters; and the Phase 1A deep reading of *Sentinel for confidence-aware multi-object tracking*. No additional external experiment is claimed here.

1. Continuity from Phase 1E and why Phase 1F exists

Phase 1E established an **offline, event-aware reference** for whether a tracker experienced a loss inside the intended occlusion/reappearance interval. That evaluator could use annotation-assisted target matching and therefore was suitable for measurement, but not for deployment. Phase 1F asks the next systems question: **can the running tracker decide for itself when its selected vehicle is no longer trustworthy?**

> **Critical separation**
>
> The Phase 1E reference loss frame is a GT-assisted evaluation signal. L1 and L2 are online detectors: they may use only signals available while the tracker runs. XML boxes, target IDs, target IoU and future reappearance annotations are prohibited inputs.

1.1 Minimal inherited context

Item	Inherited Phase 1E result used in Phase 1F
Reference event outcomes	8 tracker-event runs contained an event-level reference loss; 16 did not.
Relevant hard cases	E005-E007 contain explicit loss/resumption on the stationary SUV; E012 exposes ambiguity/false continuation near scene exit.
Trackers	SAM2.1 Tiny and DAM4SAM Tiny. Both save per-frame presence, box, mask availability and runtime.
Unavailable signal	The saved confidence field is empty for both trackers.
Scientific caution	E005-E007 reuse overlapping footage of the same stationary SUV and are not independent hard moving-target examples.

1.2 Phase 1F objective

Construct a simple online loss detector first (L1), measure its strengths and failure modes, then test whether a literature-inspired reliability-state detector (L2) adds useful sensitivity without hiding its false-alarm cost. This is baseline engineering, not the proposed sequential identity-cue novelty.

1.3 Evidence conventions

Label	Meaning
Measured	Direct CSV/terminal output from Phase 1F scripts.
Derived	Metric calculated from TP/TN/FP/FN or measured frame differences.
Diagnostic	Targeted inspection used to explain a detector failure or false alarm.
Literature-inspired	Design element adapted from Sentinel; not a reproduction of Sentinel.

2. L1 - the simple online loss detector

Origin. L1 was not copied from Sentinel or another single paper. It was intentionally constructed as a simple hand-designed baseline so that later reliability-state logic could be compared against something transparent.



Runtime-only signals: prediction presence, box validity, extreme temporal discontinuity

2.1 Runtime inputs and rule

Input	How L1 uses it
present	False or missing output = unreliable observation.
bounding box	Invalid box = unreliable observation.
temporal IoU	Compared with the previous trusted box to detect extreme discontinuity.
normalized center jump	Large normalized displacement supports extreme-discontinuity detection.
confidence / XML / target IoU	Not used. Confidence is unavailable; GT-assisted signals are prohibited.

Extreme temporal discontinuity: temporal IoU < 0.10 **AND** normalized center jump > 1.0. The observation is otherwise treated as reliable if a valid prediction exists.

State rule: one unreliable observation moves TRACKING to UNCERTAIN; three consecutive unreliable observations declare LOST. LOST is sticky because ordinary tracker output is not considered sufficient proof of identity recovery.

2.2 Why a sticky LOST state matters

> **Architecture rule**
>
> Once L1 declares LOST, a later box from the same tracker does not automatically restore the selected identity. Phase 1G and later recovery logic are responsible for candidate generation and identity verification.

3. L1 diagnostic tests and all-event results

3.1 Diagnostic events

Run	Observed L1 behavior	Interpretation
E006 / SAM2.1	1094 UNCERTAIN -> 1096 LOST; loss episode start recorded as 1094	Correctly recognizes prolonged missing output.
E006 / DAM4SAM	Same 1094 -> 1096 sequence	Same explicit-loss behavior.
E012 / SAM2.1	Remains TRACKING; no L1 loss	Smooth wrong/ambiguous continuation is invisible to the simple rule.
E012 / DAM4SAM	604 UNCERTAIN -> 606 LOST; loss episode start 604	Missing outputs make loss detectable.

3.2 Corrected 24-run evaluation

The first all-event run produced two apparent false positives on E003. Those were not true L1 errors: L1 had been searched beyond the event recovery interval, whereas the Phase 1E reference searched only the intended occlusion-to-reappearance interval. After making the comparison event-aware, E003 disappeared from the error set.

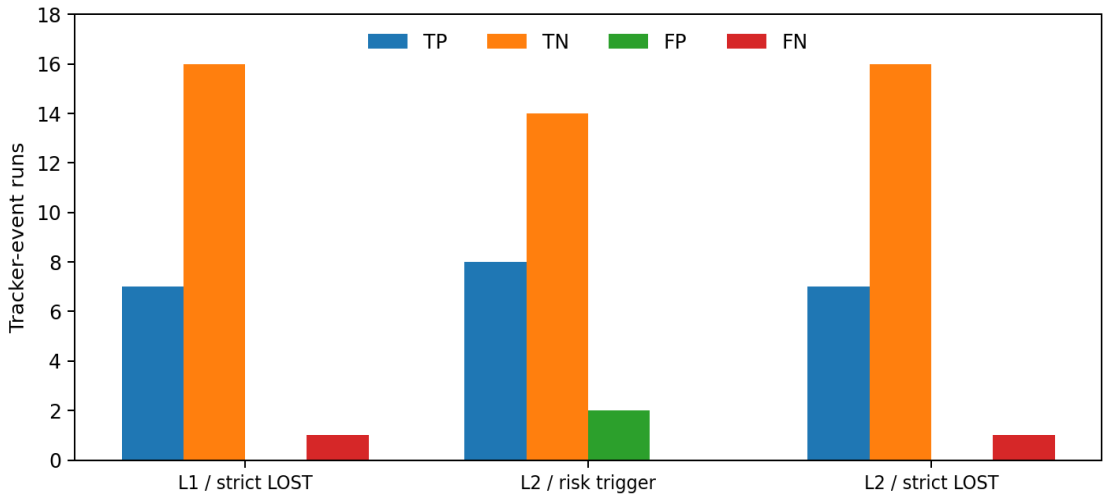


Figure 1. Confusion-count comparison used later in the L1/L2 analysis. L1 / strict LOST = TP 7, TN 16, FP 0, FN 1.

Metric	L1 result
TP / TN / FP / FN	7 / 16 / 0 / 1
Accuracy	95.8%
Precision	100.0%
Recall	87.5%
F1	93.3%
Specificity	100.0%
Mean detection delay on detected reference-loss runs	6.71 frames

4. What L1 taught us

4.1 Event-level timing on the reference-loss runs

Event / tracker	Phase 1E reference loss	L1 loss	Signed delay
E005 / SAM2.1	869	882	+13
E005 / DAM4SAM	869	881	+12
E006 / SAM2.1	1088	1094	+6
E006 / DAM4SAM	1088	1094	+6
E007 / SAM2.1	1334	1338	+4
E007 / DAM4SAM	1334	1338	+4
E012 / SAM2.1	598	None	False negative
E012 / DAM4SAM	602	604	+2

4.2 The important failure: smooth false continuation

> **Phase 1F motivation produced by L1**
>
> E012/SAM2.1 kept producing a temporally smooth box after the selected target became identity-unsafe. A rule based mainly on missing output or extreme one-step discontinuity can therefore remain confidently wrong. This is the concrete failure that motivated L2.

This finding should be carried into the final paper as a baseline limitation rather than hidden by post-hoc tuning. It also explains why **tracking availability** and **identity trustworthiness** cannot be treated as the same variable.

4.3 L1 claim that is safe to make

L1 is a conservative online loss detector. On the frozen pilot runs it produced no event-level false triggers, but missed one smooth ambiguity/false-continuation case. It is suitable as a transparent baseline and as a trigger for obvious disappearance, not as a complete identity-safety detector.

5. Sentinel basis and the L2 adaptation boundary

The Phase 1A deep reading identified Sentinel as the most relevant source for reliability-state reasoning. Sentinel assigns active trajectories to **Confident, Uncertain or Critical** states using consecutive successful hits, consecutive misses and recent mean detector confidence. Critical tracks receive additional survival/recovery processing; state also changes how motion/location and appearance evidence are weighted.

Sentinel element	What is reusable here	What is not directly transferable
State diagnosis	Progressive reliability states instead of a binary lost/not-lost decision.	Its detector-confidence signal is absent in our SAM2.1/DAM4SAM adapters.
Critical state	Represents an endangered but still-live trajectory and can trigger extra processing.	Critical is not equivalent to confirmed vehicle identity loss.
State-dependent evidence	History should change how much the system trusts current tracking evidence.	We are not reproducing Sentinel CAA/SBM or its MOT association stack.
Thresholds	Parameters should be calibrated to data.	Sentinel thresholds were dataset-specific; pedestrian values should not be copied to UA-DETRAC.

Naming discipline

L2 must be described as a **Sentinel-inspired state-based loss detector adapted to the runtime signals available from SAM2.1/DAM4SAM**. It is not an implementation or reproduction of Sentinel.

5.1 Adapter inspection

The tracker adapters confirmed that both saved outputs contain an empty confidence field. SAM2.1 propagation returns mask logits that are thresholded to build a binary mask, but the adapter explicitly records confidence=None. DAM4SAM public tracking output supplies a predicted mask and likewise records confidence=None. Pixel-level mask logits were therefore not rebranded as an identity-confidence score, and L2 was kept tracker-comparable.

6. L2 signal design and empirical threshold selection

Because detector confidence was unavailable, L2 used four online signals: **hit/stability history, miss history, temporal IoU, and normalized center jump**. The geometry thresholds were selected from the observed tails of the 24 Phase 1 runs rather than copied from Sentinel.

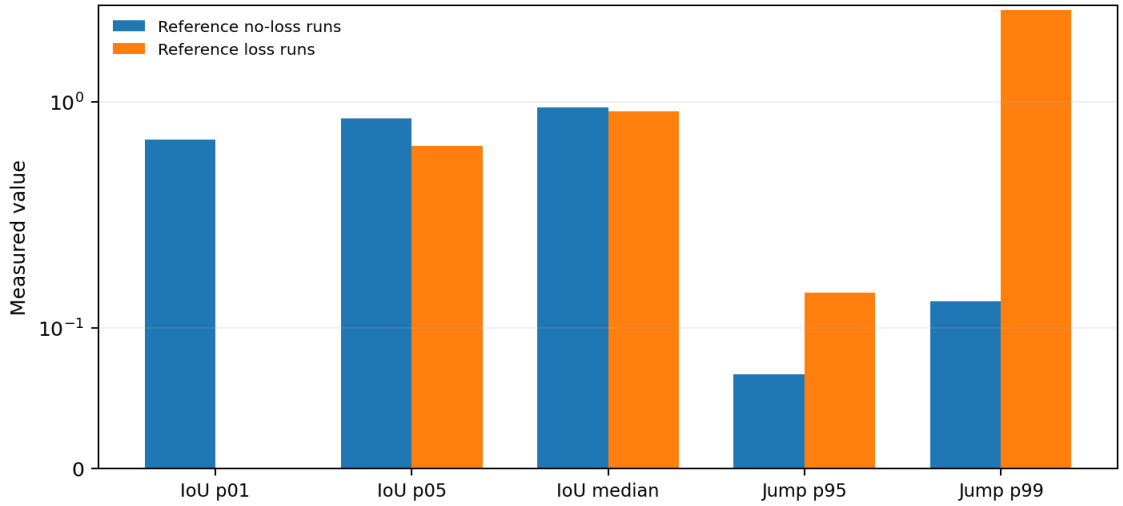


Figure 2. Distribution summary over the event-relevant intervals. The center-jump p99 on loss runs is extreme (3.075), so the chart uses a compressed logarithmic scale.

Reference group	Runs	Frames	Missing	Temporal IoU	Center jump
No loss	16	1,374	0	p01=.633; p05=.816; p10=.845; median=.933	median=.018; p90=.054; p95=.067; p99=.119
Loss	8	1,198	680	p01=.000; p05=.589; p10=.697; median=.892	median=.031; p90=.089; p95=.125; p99=3.075

6.1 Initial observation classes

Stable: temporal IoU >= 0.63 and center jump <= 0.12. **Soft anomaly:** exactly one threshold is violated. **Hard anomaly:** both thresholds are violated. Missing/invalid predictions use a separate miss path.

> **Why 0.63 and 0.12?**
>
> They approximately match the 1st-percentile temporal IoU and 99th-percentile center jump of the 16 reference no-loss runs. They are empirical pilot thresholds, not universal vehicle-tracking constants.

7. L2 design evolution: from over-aggressive to history-aware

7.1 First L2 attempt

The first state machine was CONFIDENT -> UNCERTAIN -> CRITICAL -> LOST. A hard anomaly moved directly to CRITICAL, and CRITICAL automatically became LOST after three frames even if subsequent boxes were smooth. This was intentionally tested rather than assumed correct.

Result: both E012 trackers were declared LOST from frame **541**, far before the Phase 1E reference losses (SAM2.1 598; DAM4SAM 602). The problem was not the existence of the hard anomaly; the problem was the rule that one abrupt but legitimate disturbance permanently condemned the track.

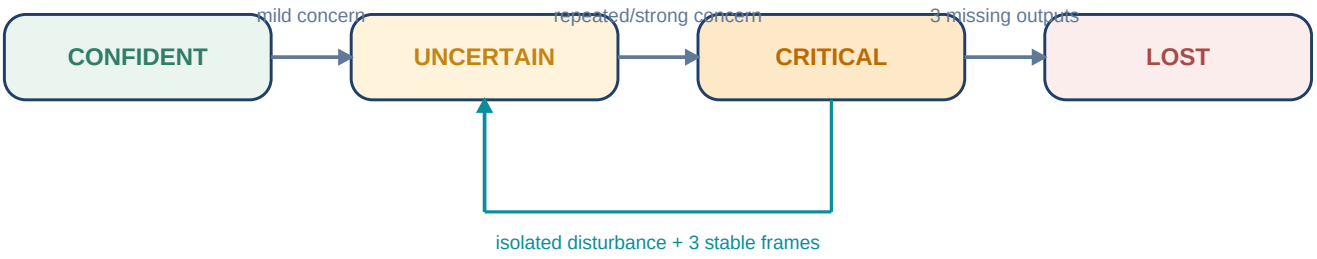
7.2 Diagnostic comparison

Region	Key pattern	Interpretation
E012/SAM2.1 541	IoU .298, jump .545, then recovery to stable behavior	Isolated severe disturbance; should be allowed to recover cautiously.
E012/SAM2.1 588-591	Hard at 588; soft at 590; hard at 591	Repeated anomaly cluster; stronger evidence that identity trust is degraded.
E004/SAM2.1 668-670	Three consecutive hard geometric anomalies	Shows that anomaly clustering alone can still occur in a no-loss event.
E004/DAM4SAM 667/669/671	Three hard anomalies interleaved with stable frames	Confirms the same geometry-only limitation in the second tracker.

7.3 Revision

The redesigned L2 introduced **hysteresis with recovery** and a **rolling anomaly history**. An isolated hard anomaly can recover from CRITICAL after three stable frames, first to UNCERTAIN and then to CONFIDENT. Two hard anomalies within a five-frame window create **persistent CRITICAL**. Missing outputs retain a separate path: first miss -> UNCERTAIN, second miss -> CRITICAL, third miss -> LOST.

8. Final L2 algorithm used for evaluation



Persistent CRITICAL is a recovery/risk trigger; LOST is reserved for explicit tracking failure.

8.1 Frozen pilot parameters

Parameter	Value	Role
temporal IoU threshold	0.63	Below this contributes low-overlap evidence.
normalized center-jump threshold	0.12	Above this contributes abnormal-displacement evidence.
stable_to_confident	3	UNCERTAIN requires three stable observations to regain CONFIDENT.
critical_recovery_stable	3	Isolated CRITICAL episode requires three stable observations before cautious recovery to UNCERTAIN.
anomaly_window	5 frames	Rolling history used to count hard anomalies.
hard_anomalies_to_persistent	2	Two hard anomalies in the window create persistent CRITICAL.
misses_to_lost	3	Three consecutive missing/invalid outputs declare LOST.

8.2 Interpretation of states

State	System interpretation
CONFIDENT	Normal tracking operation; current geometry is behaving as expected.
UNCERTAIN	Mild reliability concern; do not overreact to one abnormal frame.
CRITICAL	Tracking/identity reliability is seriously questionable; candidate/recovery processing may be activated.
LOST	Stronger explicit tracking failure, typically persistent missing output; ordinary tracker output cannot auto-restore identity.

9. Revised L2 on the diagnostic events



The revised L2 lets isolated disturbances recover, but preserves suspicion after repeated hard anomalies.

9.1 E006 - explicit disappearance

For both trackers: frame 1094 -> UNCERTAIN on first miss; 1095 -> CRITICAL on second miss; 1096 -> LOST on third miss. The loss episode start is recorded as 1094. Persistent-Critical history is not needed because the missing-output evidence is sufficient.

9.2 E012 / SAM2.1 - ambiguity without explicit disappearance

The early hard anomaly at 541 no longer causes permanent failure: CRITICAL at 541 -> UNCERTAIN at 545 -> CONFIDENT at 548. Later, the anomaly cluster around 587-591 creates **persistent CRITICAL at frame 591**. No strict LOST state is reached because SAM2.1 continues emitting boxes.

9.3 E012 / DAM4SAM - ambiguity followed by explicit loss

The early 540-541 disturbance also recovers. Later hard anomalies create **persistent CRITICAL at 585**. Consecutive missing predictions from frame 604 then produce strict LOST, declared on the third miss at 606 with loss episode start 604.

> **Key conceptual result**
>
> Persistent CRITICAL and LOST are not synonyms. CRITICAL means "do not safely trust the tracker; start/prepare recovery reasoning." LOST is reserved for stronger explicit failure. E012/SAM2.1 demonstrates why this distinction is necessary.

10. L1 versus L2 on all 24 tracker-event runs

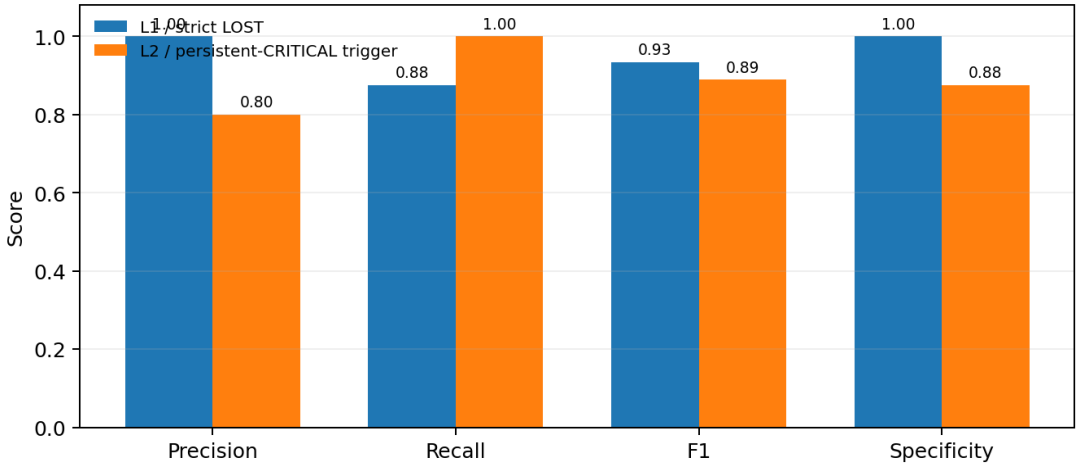


Figure 3. L1 strict-loss metrics versus the L2 persistent-Critical recovery trigger. L2 trades precision/specificity for full recall.

Output interpreted as positive	TP	TN	FP	FN	Accuracy	Precision	Recall	F1	Specificity
L1 / strict LOST	7	16	0	1	95.8%	100.0%	87.5%	93.3%	100.0%
L2 / persistent-CRITICAL risk trigger	8	14	2	0	91.7%	80.0%	100.0%	88.9%	87.5%
L2 / strict LOST	7	16	0	1	95.8%	100.0%	87.5%	93.3%	100.0%

Timing. L1 and L2 strict LOST share a mean detected-loss delay of 6.71 frames. At the L2 risk-trigger level, the mean signed trigger timing relative to the Phase 1E reference is +2.62 frames across the eight reference-loss runs; this average mixes early triggers on E012 with later explicit-loss triggers on E005-E007.

10.1 Important mismatch cases

Case	Reference	L2 persistent Critical	L2 Lost	Meaning
E004 / SAM2.1	No loss	669	None	False Critical trigger caused by legitimate severe geometry changes.
E004 / DAM4SAM	No loss	669	None	Same false-trigger pattern in the second tracker.
E012 / SAM2.1	Loss 598	591	None	L2 catches risk that L1 misses, but only as Critical, not explicit disappearance.
E012 / DAM4SAM	Loss 602	585	604	L2 raises risk early and later reaches strict Lost.

11. Failure analysis: why L2 should not be tuned to perfection

The two E004 false Critical triggers were investigated before changing parameters. Geometry alone could not cleanly separate them from the meaningful E012 ambiguity.

11.1 Motion-angle diagnostic

Sequence / tracker	Observed direction changes around anomaly	Conclusion
E004 / SAM2.1	~145°, 175°, 171° across frames 668-670	Very large motion-direction inconsistency despite no event-level loss.
E004 / DAM4SAM	~144° near 667 and additional large reversals	Same issue in a second tracker.
E012 / SAM2.1	~142°, 148°, 131° around 589-591	Also highly inconsistent, but not uniquely so.
E012 / DAM4SAM	Large reversals including ~146°, 180°, 153°	Again not separable by angle alone.

Scientific conclusion

Bounding-box geometry alone - temporal IoU, center jump and motion direction - is insufficient to reliably distinguish legitimate abrupt target motion from identity drift onto a distractor. Adding more geometric thresholds would risk overfitting these same 12 events.

11.2 Why no hidden confidence proxy was added

Adapter inspection confirmed confidence=None for both trackers. SAM2.1 mask logits are used to construct masks; DAM4SAM returns a predicted mask. These values are not equivalent to calibrated confidence that the tracker still follows the same selected physical vehicle. Introducing a SAM-specific internal proxy would also break the clean SAM2.1/DAM4SAM comparison.

11.3 Recommended interpretation

L2 is useful as a **more sensitive risk detector**, not as a perfect loss classifier. The two false Critical triggers quantify the cost of increased sensitivity. This sensitivity-vs-false-alarm trade-off should be preserved as an experimental result rather than tuned away on the pilot set.

Validity note

The 24 runs are a pilot baseline over 12 frozen UA-DETRAC events, not an independent held-out benchmark. E005-E007 overlap the same stationary target/sequence, and the L2 thresholds were derived from these pilot distributions. Future generalization claims require additional events or held-out sequences.

12. Phase 1F conclusions, artifacts and handoff

12.1 What Phase 1F established

Finding	Evidence / consequence
L1 is conservative	7 TP, 16 TN, 0 FP, 1 FN. It is reliable for explicit disappearance but misses smooth identity-unsafe continuation.
L2 adds an earlier risk state	Persistent Critical catches E012/SAM2.1, giving 8/8 reference-loss recall at the risk-trigger level.
Sensitivity has a cost	L2 risk trigger produces two false positives, both E004, reducing precision to 80%.
Critical != Lost	Critical is the point where extra recovery reasoning may be justified; Lost is a stronger explicit failure state.
Geometry has a ceiling	IoU, displacement and direction cannot by themselves resolve legitimate abrupt motion versus wrong-identity continuation.
Do not over-tune	Thresholds are pilot-calibrated; further tuning on the same events would weaken later claims.

12.2 Safe paper-ready claims

Suggested results wording

A simple online loss rule achieved 95.8% event-level accuracy with no false triggers but missed one smooth ambiguity case. A Sentinel-inspired reliability-state trigger recovered that missed case and reached 100% recall on the eight reference-loss runs, while introducing two false Critical triggers. The experiment therefore supports separating an early "Critical" recovery trigger from a stricter "Lost" state.

12.3 Implementation artifacts

Artifact	Role
src/recovery/loss_detector.py	L1 runtime state machine and temporal-discontinuity helpers.
src/evaluation/test_l1_loss_detector.py	Compact E006/E012 diagnostic test.
src/evaluation/evaluate_l1_all_events.py	Event-aware 24-run L1 evaluation; writes phase1f_l1_results.csv.
src/evaluation/analyze_l2_signals.py	Measured no-loss/loss temporal-IoU and center-jump distributions.
src/recovery/l2_loss_detector.py	Final Sentinel-inspired state detector with recovery hysteresis and rolling anomaly history.
src/evaluation/test_l2_loss_detector.py	Compact diagnostic transitions and persistent-Critical/Lost frames.
src/evaluation/evaluate_l2_all_events.py	24-run L2 risk-trigger and strict-Lost evaluation; writes phase1f_l2_results.csv.

12.4 Handoff to Phase 1G

Phase 1G can now use the Phase 1F state distinction deliberately: **CRITICAL** is an appropriate point to activate or prepare candidate generation because tracker identity is no longer safely trusted, while **LOST** represents a stronger explicit failure. Candidate generation must still be evaluated separately in real detector mode and oracle/debug annotation mode.

Appendix A. Compact event-level Phase 1F audit

Event	SAM2.1 L1	DAM4SAM L1	SAM2.1 L2 risk	DAM4SAM L2 risk	Interpretation
E001	TN	TN	TN	TN	No event-level reference loss and no persistent L2 risk trigger.
E002	TN	TN	TN	TN	No event-level reference loss and no persistent L2 risk trigger.
E003	TN	TN	TN	TN	No event-level reference loss and no persistent L2 risk trigger.
E004	TN	TN	FP @669	FP @669	No reference loss; legitimate abrupt geometry causes false Critical.
E005	TP	TP	TP	TP	Explicit loss; both detectors trigger.
E006	TP	TP	TP	TP	Explicit loss; both detectors trigger.
E007	TP	TP	TP	TP	Explicit loss; both detectors trigger.
E008	TN	TN	TN	TN	No event-level reference loss and no persistent L2 risk trigger.
E009	TN	TN	TN	TN	No event-level reference loss and no persistent L2 risk trigger.
E010	TN	TN	TN	TN	No event-level reference loss and no persistent L2 risk trigger.
E011	TN	TN	TN	TN	No event-level reference loss and no persistent L2 risk trigger.
E012	FN	TP	TP @591	TP @585	Smooth SAM ambiguity caught only by L2 risk state; DAM later Lost @604.

Appendix B. Source and interpretation notes

[P1E] Vehicle Tracking + Occlusions Research Project, Phase 1D and Phase 1E Research Documentation, 05 October 2026. Used as the immediate continuity record and source of the event-aware reference loss/recovery framework.

[S] H.-S. Yang, S.-W. Park, C.-B. Sim, S.-H. Jung, *Sentinel for confidence-aware multi-object tracking*, Scientific Reports 16, article 13571 (2026), DOI 10.1038/s41598-026-43938-2. Phase 1F adapts the reliability-state concept; it does not reproduce Sentinel.

[R] Project-generated runtime artifacts: phase1f_l1_results.csv, phase1f_l2_results.csv, Phase 1E final metrics, prediction CSVs, targeted temporal-IoU/center-jump/motion-angle diagnostics, and tracker adapter source inspection.